

Power dissipated by a conductor wire

A homogeneous filiform conductor having a cross section $S = 10^{-5} \text{ m}^2$ has a resistance $R = 2 \text{ } \Omega$. The density of the charge carriers (electrons) is worth $n = 10^{28} \text{ m}^{-3}$ and the drift velocity is $v_d = 2 \cdot 10^{-3} \text{ cm/s}$. How much is the power dissipated by the conductor wire?

- a. $P = 0.20 \text{ W}$
- b. $P = 0.64 \text{ W}$
- c. $P = 0.32 \text{ W}$
- d. $P = 0.10 \text{ W}$

Displacement current density in a capacitor

The charge q on the surface armatures S of an empty plain capacitor varies as a function of time with the law $q(t) = at^4 + b$ (where a and b are constants of suitable size). Calculate the displacement current density J_s that flows across the capacitor.

a. $J_s = at^4/S$

b. $J_s = 4at^3/S$

c. $J_s = \epsilon_0 a/S$

d. $J_s = a/S$

e. $J_s = \epsilon_0 at/S$

Drift velocity of electrons in a conductor

Calculate the modulus of the drift velocity of the carriers (electrons) that move in a conductor of circular cross section of diameter $d = 2$ mm in which flows a current $i = 12$ A. (Assume that the number of carriers per unit volume is equal to $n = 2 \cdot 10^{28} \text{ m}^{-3}$)

a. $v_d = 1.19 \cdot 10^{-3} \text{ m/s}$

b. $v_d = 5.96 \cdot 10^{-3} \text{ m/s}$

c. $v_d = 2.98 \cdot 10^{-4} \text{ m/s}$

d. $v_d = 5.96 \cdot 10^{-4} \text{ m/s}$

Resistivity of a conductor

The resistivity of a conductor depends on:

- a) the intensity of current flowing in the conductor
- b) the geometric dimensions of the conductor
- c) from the electric field that is present inside the conductor
- ☒ d) the temperature at which the conductor is located

Displacement current density formula

Which of the following formulas links the shift current density $\mathbf{j}_s$ to the electric field?

a. $\mathbf{j}_s = \frac{1}{\varepsilon_0} \frac{d\mathbf{E}}{dt}$

b. $\mathbf{j}_s = \mu_0 \frac{d\mathbf{E}}{dt}$

☒ c. $\mathbf{j}_s = \varepsilon_0 \frac{d\mathbf{E}}{dt}$

d. $\mathbf{j}_s = \varepsilon_0 \mu_0 \frac{d\mathbf{E}}{dt}$

Power dissipated per unit volume in a conductor

The power dissipated per unit volume along a conductor having a resistivity $\rho = 6 \cdot 10^{-8} \, \Omega \cdot \text{m}$ and travelled by a current density $J = 4 \cdot 10^5 \, \text{A/m}^2$ is:

a. $P_v = 9.6 \cdot 10^2 \, \text{W/m}^3$

☒ b. $P_v = 9.6 \cdot 10^3 \, \text{W/m}^3$

c. $P_v = 9.6 \cdot 10^3 \, \text{W/m}^2$

d. $P_v = 4.3 \cdot 10^3 \, \text{W/m}^3$

Drift velocity of electron carriers

Calculate the drift velocity of the carriers (electrons) that move in a conductor of circular cross section of diameter $d = 2 \text{ mm}$ in which flows a current $i = 3 \text{ A}$. (Assume that the number of carriers per unit volume is equal to $n = 2 \cdot 10^{28} \text{ m}^{-3}$)

a. $v_d = 2.98 \cdot 10^{-6} \text{ m/s}$

b. $v_d = 5.96 \cdot 10^{-4} \text{ m/s}$

☒ c. $v_d = 2.98 \cdot 10^{-4} \text{ m/s}$

d. $v_d = 2.98 \cdot 10^{-3} \text{ m/s}$

Motion of a particle in a time-dependent vector potential

In a region of space there is a potential vector $\mathbf{A} = 4azt\mathbf{j}$, where a is a constant of appropriate size. What is the system of differential equations, which, with appropriate boundary conditions, describes the trajectory of a point particle of charge q and mass m ? (Neglecting the force of gravity)

a. $\frac{d^2x(t)}{dt^2} = 0; \quad \frac{d^2y(t)}{dt^2} = \frac{4aq}{m} \left[t \frac{dy(t)}{dt} + z(t) \right]; \quad \frac{d^2z(t)}{dt^2} = \frac{4aq}{m} t \frac{dy(t)}{dt}$

b. $\frac{d^2x(t)}{dt^2} = 0; \quad \frac{d^2y(t)}{dt^2} = \frac{4aq}{m} \left[t \frac{dy(t)}{dt} - z(t) \right]; \quad \frac{d^2z(t)}{dt^2} = \frac{4aq}{m} t \frac{dy(t)}{dt}$

c. $\frac{d^2x(t)}{dt^2} = 0; \quad \frac{d^2y(t)}{dt^2} = \frac{4aq}{m} \left[t \frac{dy(t)}{dt} + z(t) \right]; \quad \frac{d^2z(t)}{dt^2} = -\frac{4aq}{m} t \frac{dy(t)}{dt}$

d. $\frac{d^2x(t)}{dt^2} = 0; \quad \frac{d^2y(t)}{dt^2} = \frac{8aq}{m} \left[t \frac{dy(t)}{dt} - z(t) \right]; \quad \frac{d^2z(t)}{dt^2} = \frac{8aq}{m} t \frac{dy(t)}{dt}$

Correct Answer: $\frac{d^2x}{dt^2} = 0; \quad \frac{d^2y}{dt^2} = -\frac{4aq}{m} \left(t \frac{dz}{dt} + z \right); \quad \frac{d^2z}{dt^2} = \frac{4aq}{m} t \frac{dy}{dt}$

Fields from Vector Potential

The vector potential $\mathbf{A}(x, y, z, t) = 4ay\mathbf{j} - bt\mathbf{k}$, where a and b are constants of suitable size, produces a magnetic field $\mathbf{B}$ and an electric field $\mathbf{E}$ equal to:

- a. $\mathbf{B} = 4a\mathbf{k}; \mathbf{E} = b\mathbf{j}$
- b. $\mathbf{B} = -4a\mathbf{k}; \mathbf{E} = 0$
- ☒ c. $\mathbf{B} = -4a\mathbf{k}; \mathbf{E} = b\mathbf{i}$
- d. $\mathbf{B} = 4a\mathbf{k}; \mathbf{E} = 0$

Correct answer for the given potential text: $\mathbf{B} = 0, \mathbf{E} = b\mathbf{k}$. (Note: Choice c is correct if $\mathbf{A} = (4ay - bt)\mathbf{i}$ as indicated in the correction note.)

Conductivity in the Drude model

The electrical conductivity σ in the Drude model

- a. doesn't depend from the sign of carriers,
- b. doesn't depend from the mean time interval between two successive collision
- c. depends from the square of the density of carriers.
- d. depends from the electric field.

Capacity of discharging capacitor

In a circuit, a capacitor C and a resistor $R = 2\,\Omega$ are connected in series with each other. Initially the circuit is open and a q_0 charge is present on the capacitor plates. If after a time $t = 10^{-5}\,\text{s}$ from the closing of the circuit the charge on the capacitor plates is halved, what is its capacity?

a. $C = 1.4 \cdot 10^{-5}\,\text{F}$

☒ b. $C = 7.2 \cdot 10^{-6}\,\text{F}$

c. $C = 2.9 \cdot 10^{-5}\,\text{F}$

d. $C = 3.6 \cdot 10^{-6}\,\text{F}$

Motion of a particle in electromagnetic fields

In a region of space there is an electric field $\mathbf{E} = 4abz\mathbf{i}$ and a magnetic field $\mathbf{B} = 2b\mathbf{j}$ where a and b are constants of suitable size. What is the system of differential equations which, with suitable boundary conditions, describes the trajectory of a point particle of charge q and mass m ? (Neglecting gravity)

a. $\frac{d^2x(t)}{dt^2} = 0; \quad \frac{d^2y(t)}{dt^2} = 4a(q/m)z(t); \quad \frac{d^2z(t)}{dt^2} = 0$

b. $\frac{d^2x(t)}{dt^2} = 2b(q/m) \left[2az(t) - \frac{dz(t)}{dt} \right]; \quad \frac{d^2y(t)}{dt^2} = 0; \quad \frac{d^2z(t)}{dt^2} = 2b(q/m) \frac{dx(t)}{dt}$

c. $\frac{d^2x(t)}{dt^2} = -2b(q/m)z(t) \frac{dx(t)}{dt}; \quad \frac{d^2y(t)}{dt^2} = 4ab(q/m)z(t); \quad \frac{d^2z(t)}{dt^2} = 2b(q/m)x(t) \frac{dx(t)}{dt}$

d. $\frac{d^2x(t)}{dt^2} = 0; \quad \frac{d^2y(t)}{dt^2} = 2(q/m)(2az(t) + b\dot{x}(t)); \quad \frac{d^2z(t)}{dt^2} = 0$

Drift velocity calculation

Calculate the drift velocity v_d of the conduction electrons inside a homogeneous metal wire of section $S = 6 \text{ mm}^2$, crossed by a current $I = 2.5 \text{ A}$. (Assume that the density of the carries n is equal to $3 \cdot 10^{28} \text{ m}^{-3}$)

a. $v_d = 3.47 \cdot 10^{-5} \text{ m/s}$

b. $v_d = 1.74 \cdot 10^{-3} \text{ m/s}$

c. $v_d = 1.74 \cdot 10^{-4} \text{ m/s}$

d. $v_d = 8.70 \cdot 10^{-5} \text{ m/s}$

Trajectory of a particle in a vector potential

In a region of space there exists a vector potential $\mathbf{A}(x, y, z, t) = A_0 x t \mathbf{i}$. What is the system of differential equations that, with appropriate boundary conditions, describes the trajectory of a particle with point charge q and mass m ? (Neglect gravity)

a. $\frac{d^2 x}{dt^2} = -\frac{q}{m} A_0 x$, $\frac{d^2 y}{dt^2} = 0$, $\frac{d^2 z}{dt^2} = \frac{q}{m} A_0 x$

b. $\frac{d^2 x}{dt^2} = -\frac{q}{m} A_0 x$, $\frac{d^2 y}{dt^2} = 0$, $\frac{d^2 z}{dt^2} = 0$

c. $\frac{d^2 x}{dt^2} = \frac{q}{m} A_0 x$, $\frac{d^2 y}{dt^2} = \frac{q}{m} A_0 t$, $\frac{d^2 z}{dt^2} = \frac{q}{m} A_0 t$

d. $\frac{d^2 x}{dt^2} = -\frac{q}{m} A_0 x$, $\frac{d^2 y}{dt^2} = -\frac{q}{m} A_0 t$, $\frac{d^2 z}{dt^2} = -\frac{q}{m} A_0 t$

e. $\frac{d^2 x}{dt^2} = 0$, $\frac{d^2 y}{dt^2} = 0$, $\frac{d^2 z}{dt^2} = 0$

Magnetic energy density from vector potential

In a region of empty space there exists a potential vector $\mathbf{A}(x, y, z, t) = ax\mathbf{i} + by\mathbf{j} + czx\mathbf{k}$ (where a, b , and c are constants of suitable size). How much is the associated magnetic density of energy w_B ?

a. $w_B = [a^2x^2 + b^2y^2 + t^2c^2x^2]/2\mu_0$

b. $w_B = 0$

c. $w_B = \mu_0[a^2x^2 + b^2y^2 + t^2c^2x^2]/2$

d. $w_B = \mu_0c^2z^2/2$

☒ e. $w_B = c^2z^2/2\mu_0$

Electric field from scalar and vector potentials

In a region of empty space there exists a potential vector $\mathbf{A}(x, y, z, t) = atx\mathbf{u}_x + by\mathbf{u}_y + cz\mathbf{u}_z$ and an electric potential $V(x, y, z) = -fx$ (where a, b, c and f are constants of suitable size). Which is the associated electric field $\mathbf{E}$?

a. $\mathbf{E} = -(f + ax)\mathbf{u}_x$

☒ b. $\mathbf{E} = (f - ax)\mathbf{u}_x$

c. $\mathbf{E} = -(f + at)\mathbf{u}_x - b\mathbf{u}_y - c\mathbf{u}_z$

d. $\mathbf{E} = -f\mathbf{u}_x$

Electromagnetic fields from potentials

In a region of space there is the vector potential $\mathbf{A} = 3aytk$ and the electric potential $V = 2bx$. What are the magnetic field $\mathbf{B}$, the electric field $\mathbf{E}$ and the displacement current density $\mathbf{j}_s$? (a and b are constant in appropriate units of measurement)

a. $\mathbf{B} = -3at\mathbf{j}, \mathbf{E} = -2b\mathbf{i} - 3ay\mathbf{k}, \mathbf{j}_s = \mathbf{0}$

b. $\mathbf{B} = 3at\mathbf{i}, \mathbf{E} = -2b\mathbf{i} - 3ay\mathbf{k}, \mathbf{j}_s = \mathbf{0}$

c. $\mathbf{B} = 3at\mathbf{i}, \mathbf{E} = -2b\mathbf{i} - 3aytk, \mathbf{j}_s = \varepsilon_0 3ay\mathbf{k}$

d. $\mathbf{B} = 3ay\mathbf{i}, \mathbf{E} = -2b\mathbf{i} - 3aytk, \mathbf{j}_s = \varepsilon_0 3ay\mathbf{k}$

Meaning of solving Maxwell's equations

Solving Maxwell's equations of electromagnetism means

- a. to calculate the fields $\mathbf{E}$ and $\mathbf{B}$, knowing the distributions of the charge densities ρ , of the current densities $\mathbf{j}$ and the boundary conditions
- b. to calculate the charge distributions Q and current distribution $\mathbf{j}$, knowing the fields $\mathbf{E}$ and $\mathbf{B}$
- c. to calculate the distribution of charge densities ρ and current densities $\mathbf{j}$, knowing the fields $\mathbf{E}$ and $\mathbf{B}$
- d. to satisfy the continuity equation $\nabla \cdot \mathbf{j} + \frac{\partial \rho}{\partial t} = 0$

Physical meaning of Poynting vector flux

The flow of the Poynting vector through a closed surface S represents

- a. the variation over time of the electromagnetic energy contained in the volume V delimited by the closed surface S
- b. the energy dissipated by the Joule effect in the volume V delimited by the closed surface S
- c. the electromagnetic energy contained in the volume V bounded by the closed surface S
- d. the amplitude of the electromagnetic wave that crosses the closed surface S

Electrodynamics

Trajectory of a Charged Particle in Non-Uniform $\mathbf{E}$ and $\mathbf{B}$ Fields (Lorentz Force Trajectory)

In a region of space there is an electric field $\mathbf{E} = -4z\mathbf{j}$ and a magnetic field $\mathbf{B} = 2x\mathbf{j}$. What is the system of differential equations which, with appropriate boundary conditions, describes the trajectory of a point charge particle q and mass m ? (Neglecting gravity)

- a. $\frac{d^2x(t)}{dt^2} = 0$; $\frac{d^2y(t)}{dt^2} = \left(\frac{q}{m}\right)(4z(t) + 2x(t))$; $\frac{d^2z(t)}{dt^2} = 0$;
- b. $\frac{d^2x(t)}{dt^2} = -8\left(\frac{q}{m}\right)x(t)\frac{dz(t)}{dt}$; $\frac{d^2y(t)}{dt^2} = 4\left(\frac{q}{m}\right)z(t)$; $\frac{d^2z(t)}{dt^2} = -8\left(\frac{q}{m}\right)x(t)z(t)$;
- c. $\frac{d^2x(t)}{dt^2} = 2\left(\frac{q}{m}\right)x(t)\frac{dz(t)}{dt}$; $\frac{d^2y(t)}{dt^2} = 4\left(\frac{q}{m}\right)z(t)$; $\frac{d^2z(t)}{dt^2} = -2\left(\frac{q}{m}\right)x(t)\frac{dx(t)}{dt}$;
- d. $\frac{d^2x(t)}{dt^2} = -2\left(\frac{q}{m}\right)x(t)\frac{dz(t)}{dt}$; $\frac{d^2y(t)}{dt^2} = -4\left(\frac{q}{m}\right)z(t)$; $\frac{d^2z(t)}{dt^2} = 2\left(\frac{q}{m}\right)x(t)\frac{dx(t)}{dt}$;

Electrodynamics

Electric Field from Scalar and Vector Potentials

In a region of empty space there exists a potential vector $\mathbf{A}(x, y, z, t) = atx\mathbf{u}_x + by\mathbf{u}_y + cz\mathbf{u}_z$ and an electric potential $V(x, y, z) = -fx$ (where a, b, c and f are constants of suitable size). Which is the associated electric field $\mathbf{E}$?

- a. $\mathbf{E} = -(f + ax)\mathbf{u}_x$
- ☒ b. $\mathbf{E} = (f - ax)\mathbf{u}_x$
- c. $\mathbf{E} = -(f + at)\mathbf{u}_x - b\mathbf{u}_y - c\mathbf{u}_z$
- d. $\mathbf{E} = -f\mathbf{u}_x$

Electrodynamics

Magnetic Field from Vector Potential

In a region of empty space there exists a potential vector $\mathbf{A}(x, y, z, t) = atx\mathbf{u}_x + by\mathbf{u}_y + cz\mathbf{u}_z$ (where a , b , and c are constants of suitable size). What is the associated magnetic field $\mathbf{B}$?

- a. $\mathbf{B} = 0$
- b. $\mathbf{B} = -at\mathbf{u}_x - b\mathbf{u}_y - c\mathbf{u}_z$
- c. $\mathbf{B} = at\mathbf{u}_x + b\mathbf{u}_y + c\mathbf{u}_z$
- ☒ d. $\mathbf{B} = -c\mathbf{u}_y$

Electrodynamics

Magnetic Field of a Moving Point Charge

6. The magnetic field generated at a point P by a moving (non-relativistic) charge
- a. is inversely proportional to the distance between the moving charge and the point P
 - b. does not depend on the direction of motion of the charge
 - ☒ c. is inversely proportional to the square of the distance between the charge and the point P
 - d. does not depend on the distance between the charge and the point P

Electrodynamics

Magnetic Field of a Moving Point Charge

6. The magnetic field generated at a point P by a moving (non-relativistic) charge
- a. is inversely proportional to the distance between the moving charge and the point P
 - b. does not depend on the direction of motion of the charge
 - ☒ c. is inversely proportional to the square of the distance between the charge and the point P
 - d. does not depend on the distance between the charge and the point P

Induced current in a square loop

A square metal coil with side $a = 10$ cm and resistance $R = 10\ \Omega$, lies in the xy plane. At time $t = 0$ s a magnetic field is applied; for $t > 0$ s the magnetic field has the following form: $\mathbf{B}(x, y, z, t) = B_0 t [(1 + 3x)\mathbf{i} + (x + y)\mathbf{j} + (1 - 4z)\mathbf{k}]$ with $B_0 = 1$ T/s. What is the modulus of the current I induced in the loop? (The x, y and z coordinates are dimensionless)

- ☒ a. $I = 0.001$ A
- b. $I = 1.0$ A
- c. $I = 10$ A
- d. $I = 0.1$ A
- e. $I = 0.01$ A

EMF induced in a coil by a solenoid

An indefinite rectilinear solenoid of radius R_1 and linear density of turns n is traversed by a current of intensity $i = at$. A circular flat coil, formed by N turns with radius $R_2 > R_1$ is arranged around the solenoid, coaxially to it. The system is in a vacuum. The electromotive force (f.e.m.) induced in the coil is

a. f.e.m. $= -\mu_0 n \pi R_2^2 a$

☒ b. f.e.m. $= -\mu_0 n N \pi R_1^2 a$

c. f.e.m. $= -\mu_0 n N \pi R_2^2 a$

d. f.e.m. $= -\mu_0 n \pi R_1^2 a$

Displacement current density calculation

To electric field $\mathbf{E} = at\mathbf{k}$, where a is a constant of suitable size, is associate a displacement density of current $\mathbf{j}_s$.

a. $\mathbf{j}_s = \mu_0\epsilon_0 a\mathbf{k}$

b. $\mathbf{j}_s = \mu_0 a\mathbf{k}$

c. $\mathbf{j}_s = a\mathbf{k}$

d. $\mathbf{j}_s = \epsilon_0 a\mathbf{k}$

Displacement current density in a capacitor

The charge q on the surface armatures S of an empty plain capacitor varies as a function of time with the law $q(t) = at^4 + b$ (where a and b are constants of suitable size). Calculate the displacement current density J_s that flows across the capacitor.

a. $J_s = \epsilon_0 at/S$

b. $J_s = 4at^3/S$

c. $J_s = a/S$

d. $J_s = \epsilon_0 a/S$

Resonance frequency of RLC circuit

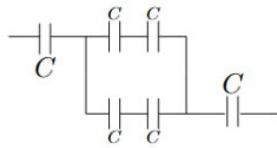
An external electromotive force $F(t) = F_0 \cos(\omega t + \phi)$ is applied to a RCL circuit in series. Knowing that $R = 0.7 \, \Omega$, $L = 8$ Henry and $C = 2 \, \mu\text{F}$ what is the value of the resonance frequency ω_0 of the circuit?

- a. The data provided are insufficient to reply.
- b. $\omega_0 = 500 \, \text{Hz}$
- ☒ c. $\omega_0 = 250 \, \text{Hz}$
- d. $\omega_0 = 2 \cdot 10^{-3} \, \text{Hz}$

Correct answer: $\omega_0 = 250 \, \text{rad/s}$

Equivalent capacitance of a capacitor network

What is the equivalent capacitance C_{eq} of the system of capacitors shown in the figure? Assume that all capacitors have the same capacity C .



a. $C_{eq} = 2C/3$

b. $C_{eq} = 4C/3$

c. $C_{eq} = C/6$

d. $C_{eq} = C/3$

Induced current in a square loop

A square metal coil with side $a = 10$ cm and resistance $R = 10\ \Omega$, lies in the xy plane. At time $t = 0$ s a magnetic field is applied; for $t > 0$ s the magnetic field has the following form:

$$\mathbf{B}(x, y, z, t) = B_0 t [(1 + 3x)\mathbf{i} + (x + y)\mathbf{j} + (1 - 4z)\mathbf{k}]$$

with $B_0 = 1$ T/s. What is the modulus of the current I induced in the loop? (The x, y and z coordinates are dimensionless)

☒ a. $I = 0.001$ A

b. $I = 1.0$ A

c. $I = 10$ A

d. $I = 0.1$ A

e. $I = 0.01$ A

Motional EMF in a rectangular loop

The rectangular loop shown in the figure has a movable arm of length l (side AB) that moves to the right with a velocity v . If the coil is immersed in a uniform magnetic field $\mathbf{B}$ directed perpendicular to the plane containing the loop, what is the magnitude of the induced electromotive force in the coil?

a. $\epsilon = 0$

b. $\epsilon = \frac{dB}{dt}lv$

c. $\epsilon = Bl\frac{dv}{dt}$

d. $\epsilon = Blv$

Time derivative of charge density

A region of a conductor is crossed by a current having density equal to $\mathbf{J}(x, y, z, t) = aty\mathbf{u}_x - by\mathbf{u}_y + cz\mathbf{u}_z$ (where a, b , and c are constants of suitable size). How much is the time derivative of the charge density, $d\rho/dt$, present in this area?

- a. $d\rho/dt = at - b + c$
- b. $d\rho/dt = c - b$
- ☒ c. $d\rho/dt = b - c$
- d. $d\rho/dt = b$

Displacement current in a receding capacitor

The plates of a flat, charged and isolated capacitor recede over time. In the hypothesis of neglecting the edge effects, the displacement current density inside the capacitor:

- a. increases over time
- b. decreases over time
- c. is constant over time but not necessarily null
- d. increases linearly over time
- ☒ e. is null

Shift current density formula

Which of the following formulas links the shift current density $\mathbf{j}_s$ to the electric field?

a. $\mathbf{j}_s = \frac{1}{c_0} \frac{d\mathbf{E}}{dt}$

b. $\mathbf{j}_s = \mu_0 \frac{d\mathbf{E}}{dt}$

☒ c. $\mathbf{j}_s = \epsilon_0 \frac{d\mathbf{E}}{dt}$

d. $\mathbf{j}_s = \epsilon_0 \mu_0 \frac{d\mathbf{E}}{dt}$